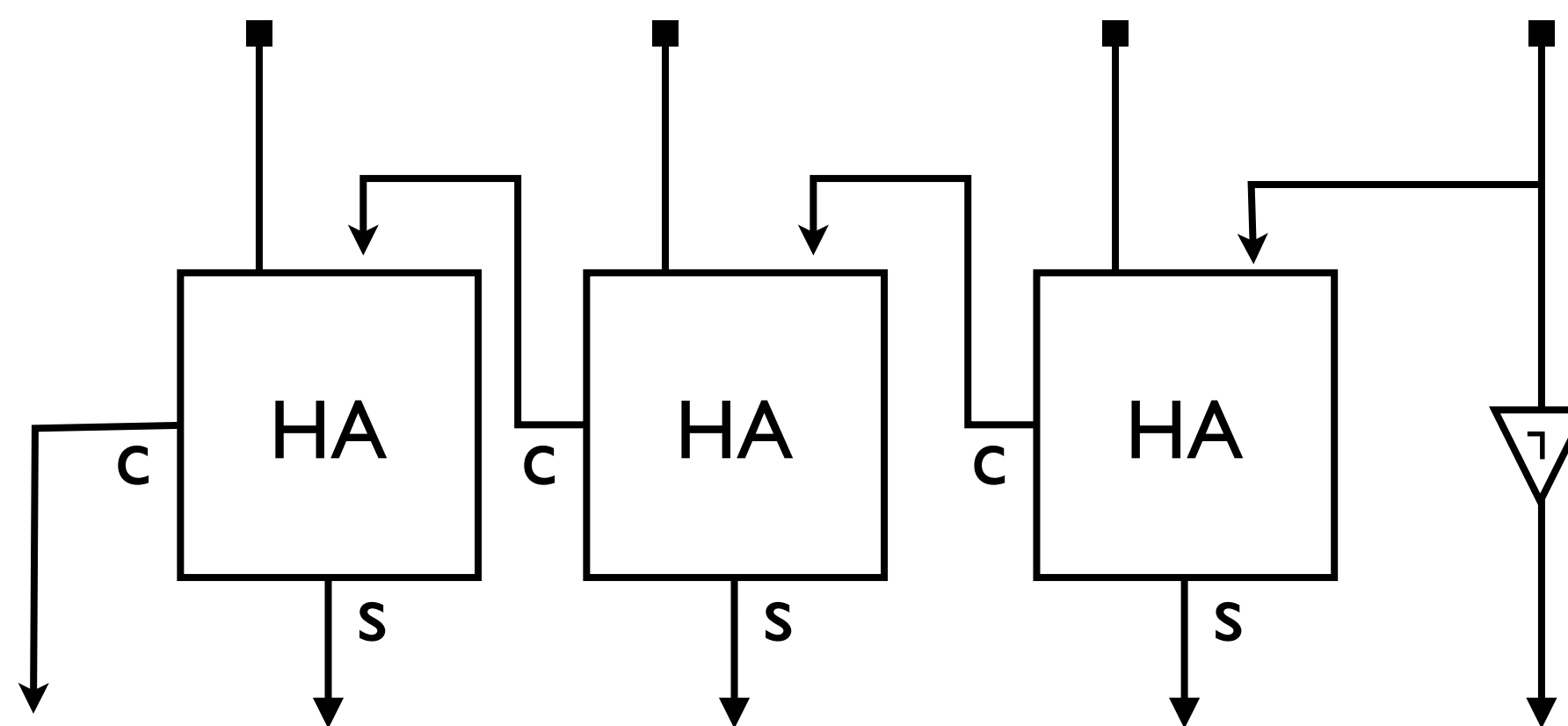


I0. Lessons from Binary Successor

The +1 B Combinatorial Circuit



+1[0-15] B combinatorial circuit

Lesson 1: The physical realization of cognition

+1B is a blueprint for the hypothesis that an entirely physical system can perform (basic) rational operations— provided we have a way of establishing representation relations between symbols and contents. A key step in our answer to Leibniz.

“[T]hree famous philosophical dilemmas stood in the way [of Hobbes’s idea]: (i) the metaphysical problem of mind interacting with matter; (ii) the theoretical problem of explaining the relevance of meanings, without appealing to a question-begging homunculus; and (iii) the methodological issue over the empirical testability (and, hence, respectability) of “mentalist” explanations. The computational idea can be seen as slicing through all three dilemmas at a stroke...” (Haugeland 1981, p. 34)

“If there are no philosophical dilemmas about chess-playing computers, then why should there be any about chess-playing people – or, indeed, about human intelligence in any other form? To put it coldly: why not suppose that people *just are* computers (and send philosophy packing)?” (Haugeland 1981, p. 36)

Lesson 2: Information processing as functional description

Hammers have the function of pounding nails, and hearts have the function of pumping blood. +1B has an *information processing* function, which goes beyond the way it manipulates the physical world. “M computes +1” is the informational-level description of the machine.

Lesson 3: The design impact of representational systems

Compare +1B and +1T machines: dramatically different physical functions, same information processing function. Machines and representational systems must be designed together, or co-evolve.

Lesson 4: The causal power of representations

The value of a function depends on the argument: the value of +1 depends on whether you apply it to 2 or to 3, for example. In a computer, these mathematical dependencies are turned into causal dependencies. In the +1 B machine, the *symbol* for 2 *causes* the *symbol* for 3, just the way the number 2 determines the number 3 for the +1 function. A central idea in CTM is that we explain the inferential connections between ideas by the causal connections between symbols.

“In effect, given an interpreted formal system with true axioms and truth-preserving rules, if you take care of the syntax, *the semantics will take care of itself.*” (Haugeland 1981, p. 44)

“An automatic formal system with an interpretation such that the semantics will take care of itself is what Daniel Dennett (1981) calls a *semantic engine*. The discovery that semantic engines are possible – that with the right kind of formal system and interpretation, a machine can handle meanings – is the basic inspiration of cognitive science and artificial intelligence.” (Haugeland 1981, p. 45)

Lesson 5: Computing a function vs. implementing a function

All things *implement* the laws of physics, but only some things *compute* the laws of physics. To compute is to *represent* a function, but not to actually conform with it. +1B can compute the laws of *adding-apples-to-a-basket*, but it does not *implement* these laws. (By the way, do the planets compute their own orbit? When humans walk, do they compute or implement their trajectory?)

Lesson 6: Computation does not require explicitly represented rules

+1B computes a function not by “following” a set of rules, which would require an inner-agent to do the following, but simply by *embodying* a set of rules in its functional architecture. A machine can compute a function without *understanding* that function in any way; all it has to do is *perform* that function. Compare: the cognitive capacities of insects and babies, but also most adult human cognition, outside of explicit reasoning, including perception, motor control, autonomic functioning, and so on.

“[The rule-applier] must be regarded as a complete little person – or *homunculus* (in Latin) – inside the head, directing the thoughts like a traffic cop. The trouble is: a theory that invokes an homunculus to *explain* thinking has begged its own question, because the homunculus itself has to think, and *that* thinking has *not* been explained.” (Haugeland 1981, p. 35)

Lesson 7: Competence vs. performance

+1B shows how a physical system can have certain cognitive *capacities*— i.e. the +1B machine has the capacity to compute +1, for a range of arguments, even if it is never actually performs this operation. It also illustrates the contrast between the capacities you have *competence* for, and your actual performance. Because of entropy, friction, and the rest, how a machine actually performs can pull apart from its competence. A physical +1B machine won't compute +1 perfectly all the time. In other words, its *performance* may be flawed. We can still say that it is a +1 machine, since this is its underlying *competence*.